

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Fall 2023
Data Structures and Algorithms (CS-212T)
Assignment # 3

Semester: 3rd
Announced Date: 11-01-24
Total Marks: 03
Instructor Name: Adeel Saeed

Batch: 2022F
Due Date: 19-01-24
Marks Obtained:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
CLO 3	Construct non linear data structures and perform its operation	PLO_4 (Design of Solutions)	C3 (Apply)

Ques#1: What is a Binary Search Tree (BST)? Create a BST for the given sequence of numbers 45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48. Traverse resultant tree in Preorder, Inorder and Postorder.

Ques#2: Insert the following sequence of numbers into a max heap and min heap: [10, 7, 15, 17, 12, 20, 6, 32]

Ques#3: Use Prim's algorithm starting at node A to compute the Minimum Spanning Tree (MST) of the following graph. For the same graph , write down the edges of the MST in the order in which Kruskal's algorithm adds them to the MST.

